

■ Automated Aquaponics System Diagnostic Report

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1. Real-time Environmental Telemetry Matrix

System Parameter	Current Simulation Value	Operational Target Status
Water Temperature	29.47 °C	Optimal: 23 - 28 °C
Water pH Level	6.99	Optimal: 6.5 - 7.5
Dissolved Oxygen (O2)	6.46 mg/L	Optimal: > 5.5 mg/L
Atmospheric Humidity	70.12 %	Optimal: 40 - 80 %
Water Tank Level	71.93 %	Optimal: > 50 %
Ammonia (NH3) Load	0.34 ppm	Optimal: < 0.25 ppm
Nitrate (NO3) Level	15.00 ppm	Optimal: 10 - 40 ppm
Hydro-Pump Flow Rate	0.96 L/min	Optimal: > 1.0 L/min

2. Autonomous Root-Cause & Actionable Diagnostics

■ [THERMAL SHOCK] High Water Temperature: 29.47 °C

- The Cause: Direct solar radiation load or cooling fan failure.
- Immediate Action Required: Force-start the inline Chiller loop and activate the biological shade actuators.

■ [BIOCHEMICAL TOXICITY] Elevated Ammonia Load: 0.34 ppm

- The Cause: Over-feeding of feedstock, unconsumed feed decomposition, or biofilter bacteria breakdown.
- Immediate Action Required: Execute an automated 20% water flush, restrict feeding calculator access for 12 cycles, and verify biofilter mesh status.

3. Fish Biomass Configuration Status

- Total Fish Count: **250 Pcs** | Net Calculated Biomass: **58.75 kg**
- Global Ecosystem Health Rating: **100/100** | Feed Efficiency Score: **70/100**